

Pavel Golikov

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PROFESSIONAL PROFILE

AI Researcher specializing in machine reasoning and alignment. Blends formal logic with rigorous low-level systems engineering to architect custom agentic frameworks and highly creative adversarial evaluations. Former Military Intelligence Operator with a Top Secret clearance, bringing a rigorous, threat-modeling mindset to AI security and model capability evaluations.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, LaTeX

ML & AI Frameworks: PyTorch, vLLM, HuggingFace, Transformers

Systems & Infrastructure: Linux/Ubuntu Server, AWS, Apache Flink, Distributed GPU Clusters

Core Competencies: Large Language Models (LLMs), Machine Reasoning, AI Alignment, Context Management, Mechanistic Interpretability, Adversarial Evaluations, Threat Modeling, Distributed Systems

FIRST-AUTHOR AI RESEARCH

- **Robust Reasoning Benchmark (RRB)** | *Under Review at a top-tier conference* 2026
First Author | University of Toronto & Vector Institute Toronto, ON
 - Designed a highly creative adversarial evaluation framework leveraging 13 deterministic textual perturbations to decouple an LLM's mechanical deciphering from its underlying mathematical logic.
 - Demonstrated that the well-known phenomenon of attention drift occurs even *within a single query's Chain-of-Thought*, empirically showing that intermediate reasoning steps pollute the dense attention mechanism. This highlights the architectural need for working memory isolation **inside** the model's Chain-of-Thought, therefore identifying the optimal **granularity of reasoning** as a critical open research problem.
 - Engineered custom mechanistic interpretability pipelines in **PyTorch** to extract and analyze causal attention probability matrices across token index boundaries, testing models ranging from 7B to 30B parameters.
 - Identified critical safety-filter vulnerabilities in **Claude 4.6 Opus**, discovering that current alignment strategies penalize abstract character-level reasoning by misclassifying inputs as prompt injections.
 - Deployed large-scale distributed inference experiments using **vLLM** across internal multi-node clusters.
- **Context Management Benchmark** | *Active Research* 2026 – Present
Lead Researcher Toronto, ON
 - Engineered a custom, tightly coupled LLM agent and evaluation engine from scratch - bypassing off-the-shelf MCP solutions - to dynamically compile cross-domain algorithmic tasks (Python/Math) into programmable computational graphs, utilizing AST parsing and dynamic rejection sampling to rigorously benchmark multi-step state tracking and working memory.
- **Fusing Adds and Shifts for Efficient Dot Products** | *IEEE CAL* 2025
First Author | Hardware ML Research Toronto, ON
 - Proposed and validated a novel algorithmic optimization for dot-product computations, demonstrating a strong foundational understanding of hardware-level ML primitives and efficiency.

ENGINEERING & PROFESSIONAL EXPERIENCE

- **Systems & Infrastructure Engineering (MSc Thesis)** 2020 – 2022
University of Toronto Toronto, ON
 - Engineered a flexible IoT distributed data-streaming framework from scratch, designed to automatically partition computational streaming queries between edge devices and cloud instances.
 - Built the full software stack: programmed Arduino/C++ sensors for real-time biological data collection (EMG/ECG), developed custom socket networking protocols, and deployed cloud infrastructure using **AWS** and **Apache Flink**.
- **Intelligence Operator** 2013 – 2018
Canadian Armed Forces Canada
 - Held a **Top Secret** security clearance, conducting rigorous analysis of classified information streams to produce actionable intelligence reports for command elements.
 - Developed a strong adversarial threat-modeling mindset, emphasizing operational security, rigorous data validation, and the identification of logical vulnerabilities in complex, multi-agent scenarios.
- **Mathematics Teacher** 2012 – 2013
Blyth Academy Toronto
 - Taught foundational mathematics, developing the ability to distill and communicate complex quantitative concepts.

EDUCATION

- **University of Toronto** Toronto, ON
PhD in Computer Science (Paused to transition to industry) 2022 – Present
- **University of Toronto** Toronto, ON
Master of Science (MSc) in Computer Science 2020 – 2022
- **University of Toronto** Toronto, ON
Bachelor of Science (BSc) in Mathematics and Philosophy (Formal Logic) Graduated 2011